

Heaven's Light of Our Guide
Rajshahi University of Engineering & Technology
Department of Mechatronics Engineering
4th Year Odd Semester Examination, 2025 (2021 Series)
Assignment

Course No.: MTE 4101

Mark: 10

Course Title: Automation

Deadline: Before the 4th Year Odd Semester Board Viva

CO- Course Outcome (CO2), **PO-** Program Outcome (PO1)

1. Title: An Automated Greenhouse System with Efficient Climate, Irrigation and Lighting Control with Fault Recovery Approach.

System Description

➤ **Stage 1: Climate Humidity Control**

- i. Temperature and humidity sensors continuously monitor two climate zones (A and B) independently.
- ii. When temperature exceeds the set-point in a zone, the roof vent motor and circulation fan for that zone activate.
- iii. If wind speed sensor or rain sensor triggers, all roof vent motors close immediately regardless of temperature state.
- iv. If air humidity in either zone exceeds the high threshold, circulation fans activate to improve air exchange.
- v. Vent motors open only if rain sensor is inactive. Fan runs for a timed duration.

➤ **Stage 2: Precision Zone Irrigation**

- i. Three irrigation zones (A, B, C) are controlled by individual solenoid valves.
- ii. Soil moisture sensors trigger irrigation for each zone independently when soil is dry.
- iii. Irrigation pump activates only when at least one zone valve is open. Pump stops when all valves close.
- iv. A pressure sensor monitors the irrigation line; low pressure triggers a fault alarm.

➤ **Stage 4: Fertilizer Dosing**

- i. Fertilizer dosing valve opens for a timed injection cycle at a configured interval.
- ii. Fertilizer tank low-level sensor triggers an alarm and inhibits dosing until tank is refilled.
- iii. Dosing occurs only when the irrigation pump is running, ensuring correct dilution.

➤ **Stage 5: Grow Lighting Control**

- i. One Lights activate in each zone when ambient light sensor detects insufficient natural light.
- ii. Lights are inhibited if zone temperature is already above the high-limit threshold.

➤ **Stage 7: Fault Detection & Recovery**

- i. Pump overload, pressure loss, fertilizer tank empty, and sensor faults trigger system fault alarm.
- ii. Fault indicator activates. Operator must resolve fault and press reset to restore operation.

Tasks

- i. Describe automation, its importance in Industry 4.0, state of automation, challenges in implementing automation in Bangladesh, and possible solutions.
- ii. Draw a complete I/O mapping table for all sensors and actuators.
- iii. Write step-by-step sequential logic for each stage.
- iv. Implement the full ladder diagram in PLC simulation software.
- v. Demonstrate the system in simulation with the following scenarios:
 - a. Normal operation (climate control, irrigation, and grow lighting all active).
 - b. Over-temperature scenario.
 - c. Soil dry scenario.
 - d. Fertilizer empty / pump fault scenario (alarm, zone pause, others continue).
- vi. Address the societal and environmental impact of precision greenhouse automation.
- vii. Write a conclusion summarizing your design, simulation results, and learning outcomes.
- viii. Provide sufficient references.